

GeoGuard

Satellite Air Quality Intelligence

Dated atmospheric measurements and AI-assisted methane screening for environmental investigation

Team GeoGuard · Theme: Air Quality Intelligence · UAE pilot

Environmental investigation

User

**Environmental monitoring
and enforcement teams**

Decision

**Where should we review
satellite evidence and
request field measurements?**

Investigation scenario

An analyst reviews an area before requesting gas-specific field observations. Separate products and missing dates complicate that review; no user interview is claimed.

GeoGuard's contribution

Existing inputs

Copernicus supplies satellite observations.

UNEP IMEO supplies the pretrained detector.

These are the foundations we use.

Proposed benefit: easier evidence assembly before field follow-up.

Review time and better decisions still need measurement with intended users.

Our implemented workflow

Quantity changes retain units, dates and coverage.

Evidence remains tied to its recorded location.

An export carries the record into analyst review.

Measurements in the PoC

Quantity	Sentinel-5P measurement	Display
NO ₂	Tropospheric column	μmol/m ²
CO	Total atmospheric column	mmol/m ²
SO ₂	Vertical column; assumed ground profile	μmol/m ²
Regional CH ₄	Bias-corrected dry-air column mixing ratio	ppb

Sentinel-2 imagery + MARS-S2L: candidate / no candidate / not assessable

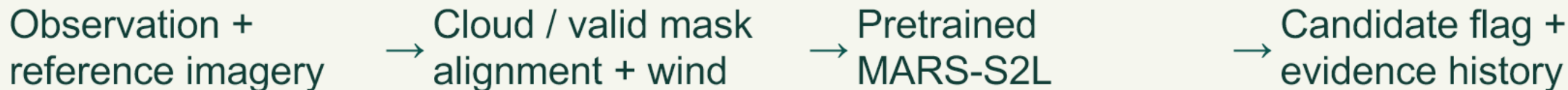
PM, VOC and H₂S instruments are not connected. Columns are not surface exposure.

Satellite evidence workflow

Atmospheric measurements



Methane screening



Different quantities and footprints remain separate. Missing observations stay unknown.

UAE atmospheric comparison

Dubai and Jebel Ali · 5 km radius comparison circles

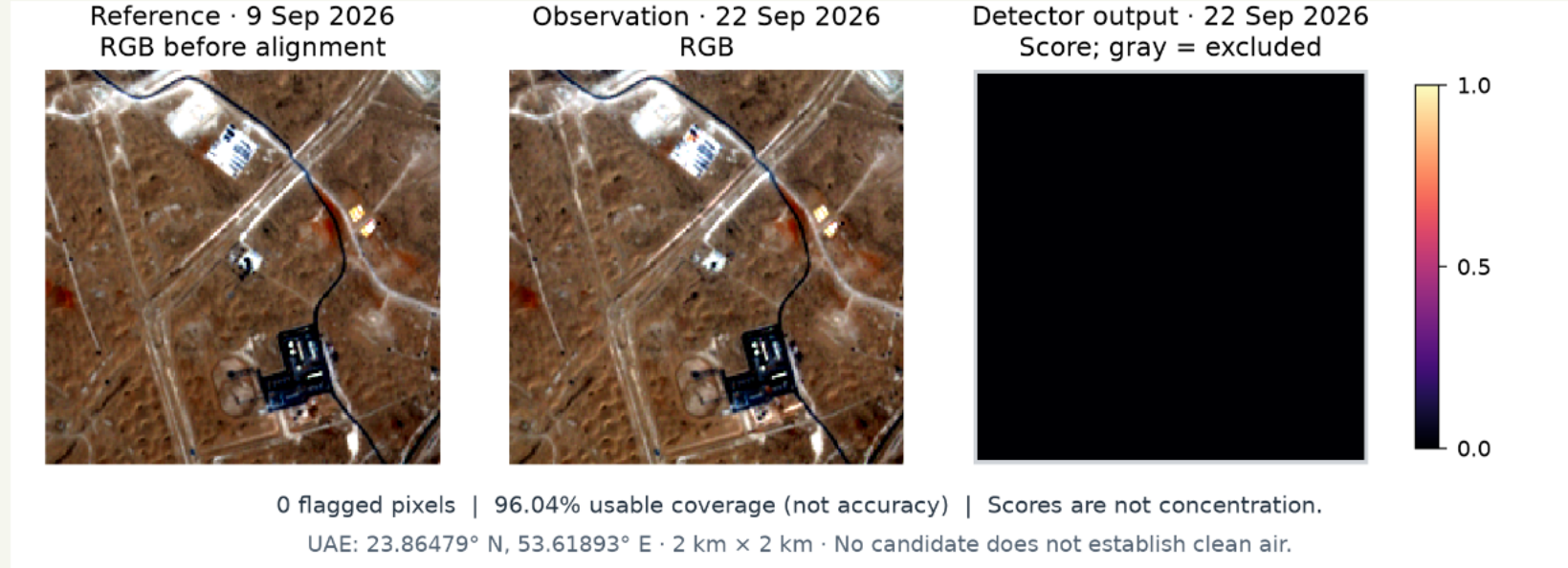
Quantity / window	Dubai mean · usable days	Jebel Ali mean · usable days
NO ₂ · Jan–Aug 2026	207.85 $\mu\text{mol}/\text{m}^2$ · 226/243	206.67 $\mu\text{mol}/\text{m}^2$ · 224/243
CO · Aug–Sep 2026	36.33 mmol/m^2 · 52/61	37.70 mmol/m^2 · 60/61
SO ₂ · Aug–Sep 2026	145.83 $\mu\text{mol}/\text{m}^2$ · 60/61	143.96 $\mu\text{mol}/\text{m}^2$ · 60/61
Regional CH ₄ · Aug–Sep 2026	1966.93 ppb · 33/61	1969.56 ppb · 37/61

Different valid days and retrieval uncertainty limit comparisons.

These means alone do not establish significance or identify an emission source.

Dated methane screening

22 September 2026 observation · 9 September reference
23.86479° N, 53.61893° E



0

candidate pixels

96.04%

usable coverage

No candidate is not proof of zero methane. Coverage is not model accuracy.

Review an area in GeoGuard

1 Choose an area

Select Dubai and NO₂: inspect the measured column and usable days.

2 Read the evidence

Compare Jebel Ali and monthly history; the mean difference is inconclusive.

3 Inspect methane candidates

Select the separate methane example; 0 candidates does not prove zero methane.

4 Export and follow up

Download current evidence; review it before gas-specific field follow-up.

Bundled NO₂ has area/history results; CO, SO₂ and regional CH₄ include archived map layers.

Actual area review

01 / Choose your area

UAE

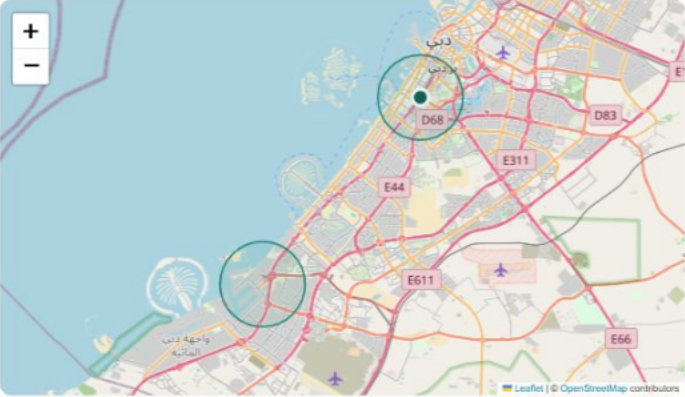
Dubai

Jebel Ali

Example methane area

+

-



Leaflet | © OpenStreetMap contributors

No raster overlay in this result. Area circles show locations; read the measured values and history.

Display scale, not a safety threshold. Transparent = missing. Pixel/time weighting can differ from the area table.

Click anywhere in the UAE study region.

Base map = navigation, not satellite measurement

Latitude

25.2048

Longitude

55.2708

Select point

Copy coordinates

Check latest satellite image

Start AI in Colab · Step 5A

Start AI in Colab, then run Step 5A. The connected website appears there while your runtime stays open.

Dubai comparison area selected.

02 / Read the evidence

EXAMPLE EVIDENCE

Satellite quantity

Nitrogen dioxide · NO₂

NO₂

NITROGEN DIOXIDE · SENTINEL-5P

207.85 $\mu\text{mol}/\text{m}^2$

Tropospheric NO₂ column over a 5 km radius area.
This is not a ground-level concentration.

2026-01-01 → 2026-09-01 (end excluded)
226 / 243 days have usable values

CH₄

METHANE SCREENING · SENTINEL-2

Not run here

Run a new methane check for this point in the notebook. The saved example belongs only to its recorded location.

Separate quantities, with separate evidence.

NO₂, CO and SO₂ are atmospheric columns; regional CH₄ is a column-averaged dry-air mixing ratio. Detailed methane screening flags patterns needing review. Neither gives a ground-level safety verdict.

Dubai · NO₂

207.85 $\mu\text{mol}/\text{m}^2$
226 / 243 usable days
January–August 2026

The methane record belongs to another selected location.

The analyst receives dated evidence, not a source verdict.

Reproducibility and validation

Reproducible data workflow

The pinned PoC recalculates archived atmospheric inputs and replays methane evidence.

Daily, monthly and period summaries match the exported observations.

30 software tests passed on 7 October 2026; outputs are available for review. Reviewed methane-positive/negative cases and false-alarm performance are not established.

Software verification does not establish independent UAE detection accuracy.

Proposed validation pilot

Scientific evaluation

One willing monitoring partner, one area;
four weeks after data access is arranged.

Review plume / no-plume and confounder
cases.

Freeze thresholds; hold out dates and
locations.

Report event counts, precision/recall and
unknown cases where labels support them.

Planned: no partner commitment or user feedback yet. Sparse imagery may extend collection.

User evaluation and gates

Three intended users compare
review/export
with their existing workflow. Record time,
completion and interpretation errors.

Agree accuracy/false-alarm targets first.
Advance only if targets are met and all
three
users interpret dates, units and unknowns.

Team and demonstration

Zainab

Pollutants, sources
and sensing methods

Raseel

Earth Engine mapping
and comparison areas

Abdulaziz

AI methane screening
and project integration

Demo and reproducible PoC

losth6wz.github.io/Geoguard/ · github.com/losth6wz/Geoguard

Data and model attribution

Copernicus Sentinel-5P / Sentinel-2

2026 satellite observations; Earth Engine and public imagery archives

UNEP IMEO MARS-S2L + CloudSEN12

Pretrained models; MARS-S2L CC BY-NC-SA 4.0, CloudSEN12 weights CC BY-NC 4.0

NASA GEOS-FP / recorded GFS fallback

Modeled wind matched to acquisition time; source and fallback recorded

Full references and applicable terms: `THIRD_PARTY_NOTICES.md` in the repository